

LAGOON CITY, ON, Canada

WMO: 712820

Lat: 44.548N

Lon: 79.221W

Elev: 221

StdP: 98.70

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.9	(c) -20.6	(d) -27.6	(e) 0.3	(f) -23.3	(g) -24.3	(h) 0.4	(i) -20.1	(j) 15.0	(k) -7.6	(l) 13.4	(m) -7.0	(n) 3.0	(o) 20	(p) 0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.4	(c) 27.4	(d) 22.6	(e) 26.1	(f) 21.7	(g) 25.0	(h) 21.1	(i) 24.1	(j) 26.0	(k) 23.0	(l) 25.0	(m) 22.0	(n) 24.2	(o) 3.7	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.7	25.3	22.3	17.4	24.4	21.2	16.3	23.6	73.6	26.1	69.3	25.0	65.5	24.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	11.2	9.8	-28.6	30.8	3.3	2.1	-31.0	32.3	-33.0	33.5	-34.8	34.7	-37.2	36.2
			-28.8	25.4	3.3	1.7	-31.1	26.6	-33.1	27.6	-34.9	28.5	-37.2	29.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-8.4	-7.4	-2.5	4.7	12.4	17.7	20.5	19.8	16.0	9.4	2.8	-3.3
	DBStd	11.20	7.01	6.06	5.82	4.25	4.11	3.44	2.89	2.78	3.78	4.26	4.66	5.56
	HDD10.0	2328	571	487	388	163	22	0	0	0	2	64	219	412
	HDD18.3	4376	829	720	646	408	189	51	12	17	90	277	467	670
	CDD10.0	1194	0	0	1	6	96	233	326	304	181	46	2	0
	CDD18.3	199	0	0	0	0	5	33	79	63	18	1	0	0
	CDH23.3	738	0	0	0	1	14	121	336	216	50	1	0	0
	CDH26.7	77	0	0	0	0	1	12	41	19	3	0	0	0
Wind	WSAvg	4.5	4.9	4.6	4.4	4.6	4.1	3.9	4.0	3.9	4.1	5.0	5.4	5.4
Precipitation	PrecAvg	960	88	61	59	68	93	77	88	70	84	83	92	91
	PrecMax	1121	169	107	119	137	154	122	135	114	130	141	182	139
	PrecMin	829	43	23	18	22	42	30	24	23	29	20	47	42
	PrecStd	74	33	23	30	29	27	26	31	22	29	37	38	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	6.5	12.8	18.6	25.2	28.4	29.7	28.9	27.0	21.6	13.9	9.1
		MCWB	6.1	5.2	9.4	12.2	19.9	22.5	24.7	23.6	21.6	18.5	11.7	7.8
	2%	DB	3.8	4.1	8.7	14.7	22.6	26.2	27.7	26.7	24.7	19.2	12.2	6.9
		MCWB	3.2	3.1	5.9	10.0	18.4	21.7	22.9	22.4	20.9	16.7	10.5	6.2
	5%	DB	2.3	2.5	6.4	12.7	20.6	24.7	26.3	25.5	23.2	17.6	10.7	5.1
		MCWB	1.7	1.6	4.6	8.9	16.6	20.6	21.9	21.4	20.1	15.4	9.1	4.2
	10%	DB	1.0	1.1	4.7	10.7	18.5	23.1	25.2	24.4	21.9	15.7	9.3	3.5
		MCWB	0.4	0.4	3.1	7.6	15.0	19.5	21.2	20.7	19.0	13.6	8.1	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	5.6	10.7	13.7	21.6	24.2	26.1	25.3	23.1	19.6	12.5	8.4
		MCDB	6.4	6.2	11.7	17.3	23.4	26.9	28.5	27.0	24.9	21.0	13.5	9.0
	2%	WB	3.4	3.2	6.7	11.4	19.5	22.7	24.4	23.7	21.7	17.3	10.9	6.2
		MCDB	3.6	3.8	7.9	13.3	22.1	25.0	26.4	25.4	23.9	18.7	11.9	6.9
	5%	WB	1.7	1.7	4.9	9.6	17.3	21.5	23.2	22.6	20.7	15.6	9.4	4.4
		MCDB	2.1	2.4	6.0	11.7	19.6	23.8	25.2	24.6	22.8	17.3	10.4	5.1
	10%	WB	0.5	0.4	3.3	8.0	15.6	20.3	22.1	21.5	19.5	14.0	8.2	2.7
		MCDB	1.0	1.0	4.5	10.3	17.8	22.5	24.3	23.7	21.4	15.4	9.2	3.4
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.8	8.4	7.7	7.9	7.7	7.4	7.4	8.0	6.6	6.0	6.0
		MCDBR	6.4	8.3	9.3	10.9	10.3	9.4	8.3	8.3	9.1	8.5	7.6	5.7
		MCWBR	6.3	7.5	7.1	6.8	6.0	5.6	4.9	4.9	5.7	6.4	6.7	5.4
	5% WB	MCDBR	6.5	8.2	8.8	9.4	8.9	8.2	7.1	6.9	7.6	7.8	7.1	5.8
		MCWBR	6.5	7.7	7.4	7.2	6.1	5.5	4.9	5.0	5.6	6.5	6.7	5.8
Clear-Sky Solar Irradiance	taub	0.295	0.302	0.322	0.363	0.393	0.416	0.418	0.407	0.373	0.350	0.313	0.288	
	taud	2.356	2.329	2.398	2.348	2.310	2.267	2.282	2.300	2.390	2.441	2.496	2.439	
	Ebn at Noon	821	885	915	899	878	855	849	844	848	820	802	799	
	Edh at Noon	77	95	102	117	126	132	129	122	103	86	68	65	
All-Sky Solar Radiation	RadAvg	1.39	2.26	3.58	4.47	5.37	5.83	5.92	5.12	4.05	2.35	1.42	1.03	
	RadStd	0.16	0.27	0.37	0.45	0.56	0.45	0.41	0.27	0.39	0.24	0.19	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page